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System Application dept.

	Title: ACH512 V2_EVB
	Name: Cover Page
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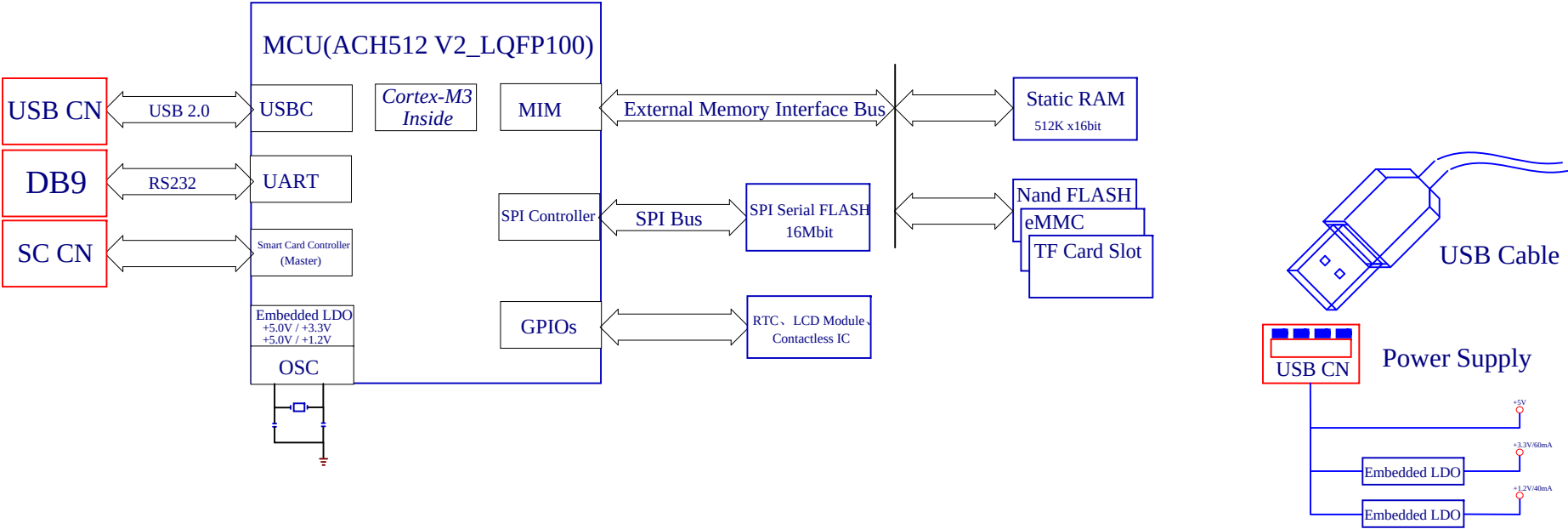
Revision History

Revision Number	Description	Revision Date
0.1	• Initial schematic	June 14,2017

 Shanghai AisinoChip Electronics Technology Co., Ltd.	Title: ACH512 V2_EVB
	Name: Content
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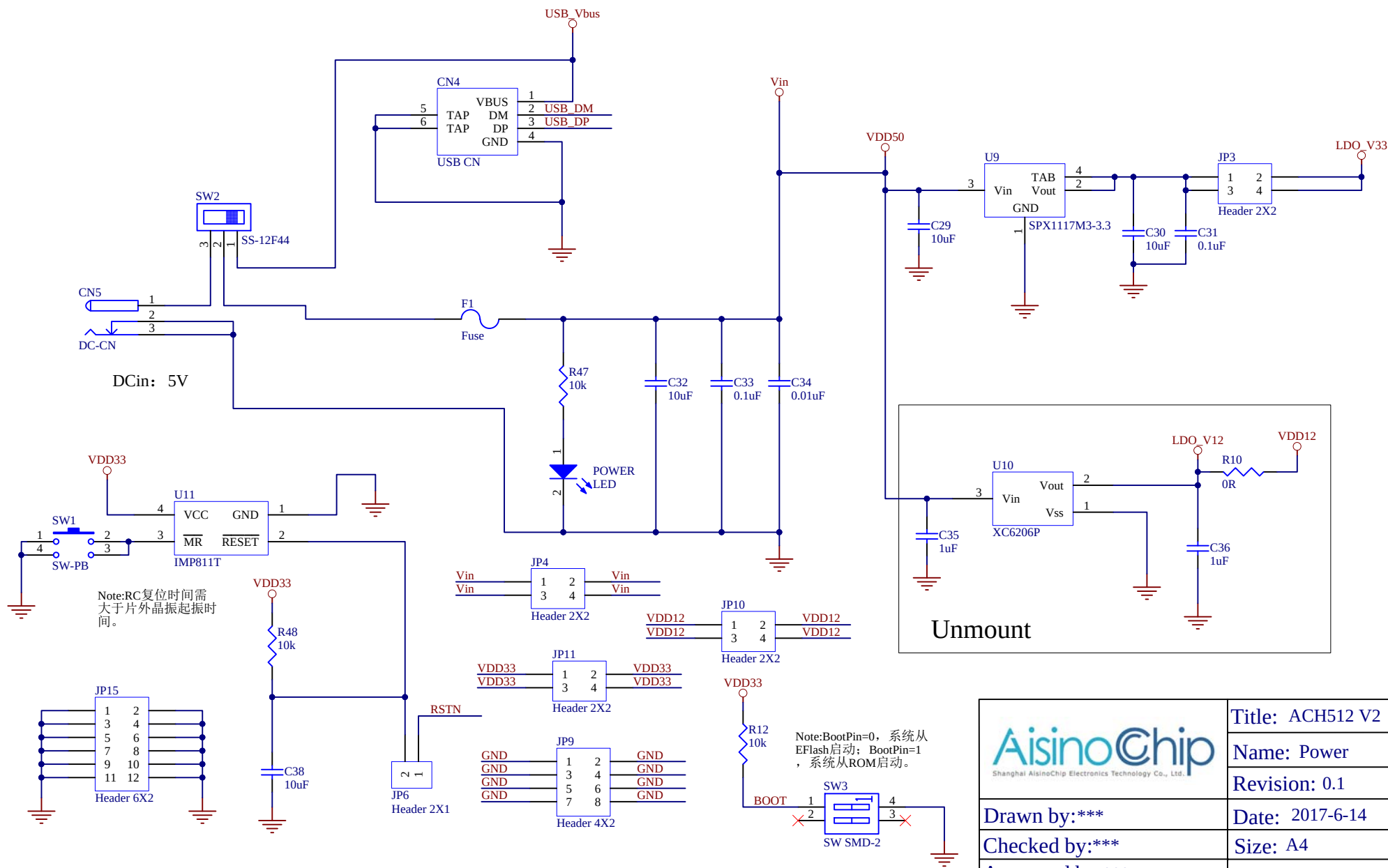
ACH512 V2_LQFP100_EVB

FUNCTIONAL DIAGRAM

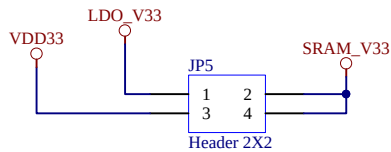


	Signal Direction
	IC or Moudle
	Connector or Switch
	Power rail
	Bus
	Digital Signal
	Analog Signal

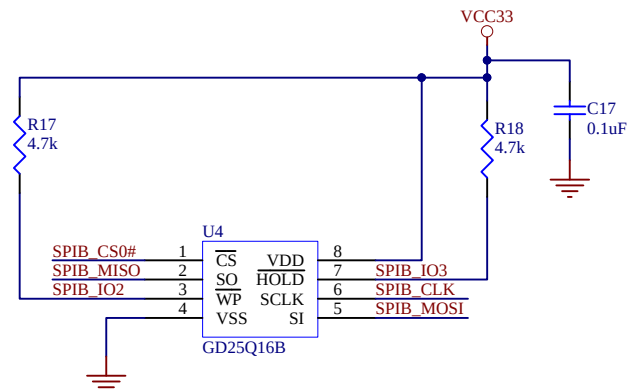
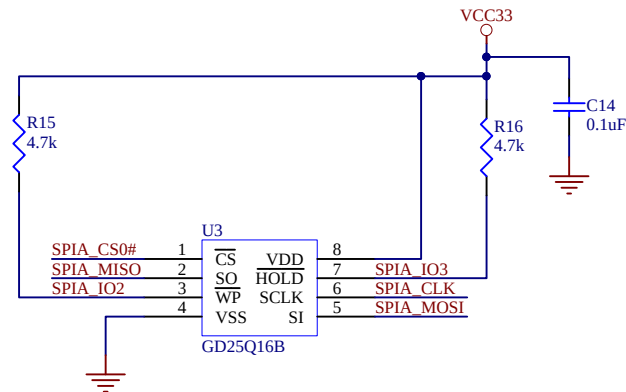
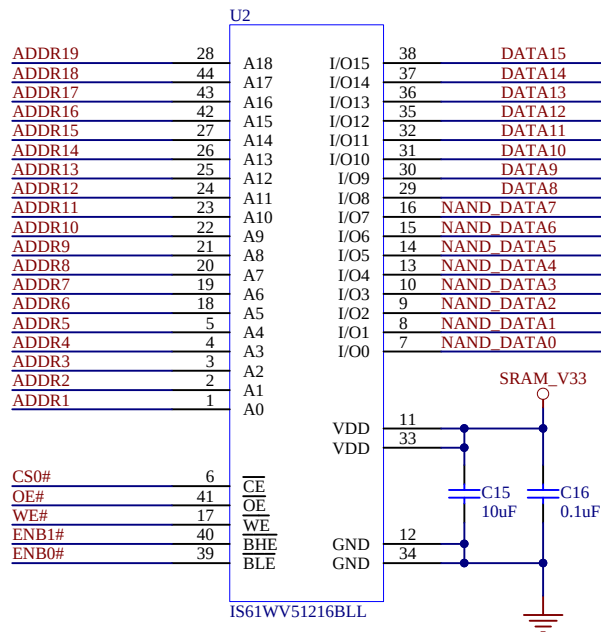
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	Name: Functional Diagram
	Revision: 0.1
Drawn by:***	Date: 2017-6-14
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Approved by:***	Sheet: 3/10



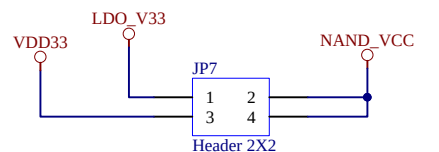
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	Name: Power
	Revision: 0.1
	Date: 2017-6-14
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Checked by:***	Sheet: 4/10
Approved by:***	



在访问片外Sram时，SPIB总线以及SPIA_IO2、SPIA_IO3信号线无法使用。

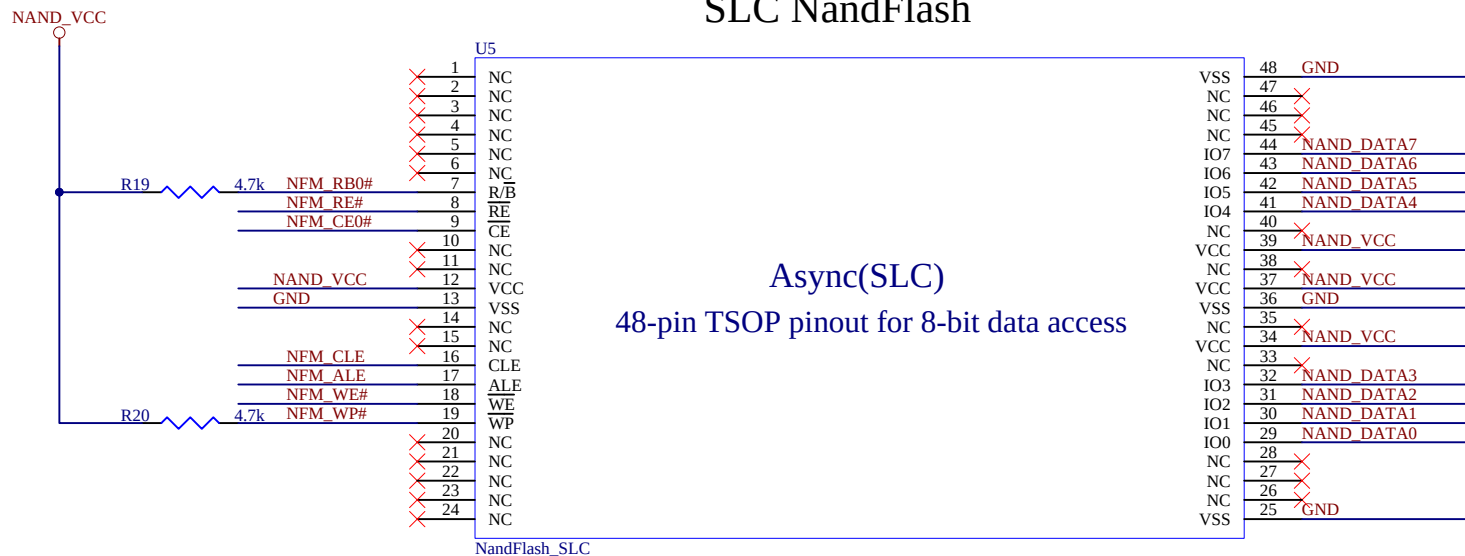


 Shanghai AisinoChip Electronics Technology Co., Ltd.	Title: ACH512 V2
	Name: Sram,SPI Flash
	Revision: 0.1
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Approved by:***	



Note: ACH512 V2支持NandFlash、eMMC和TF卡三种存储器件，实际使用时三选一。

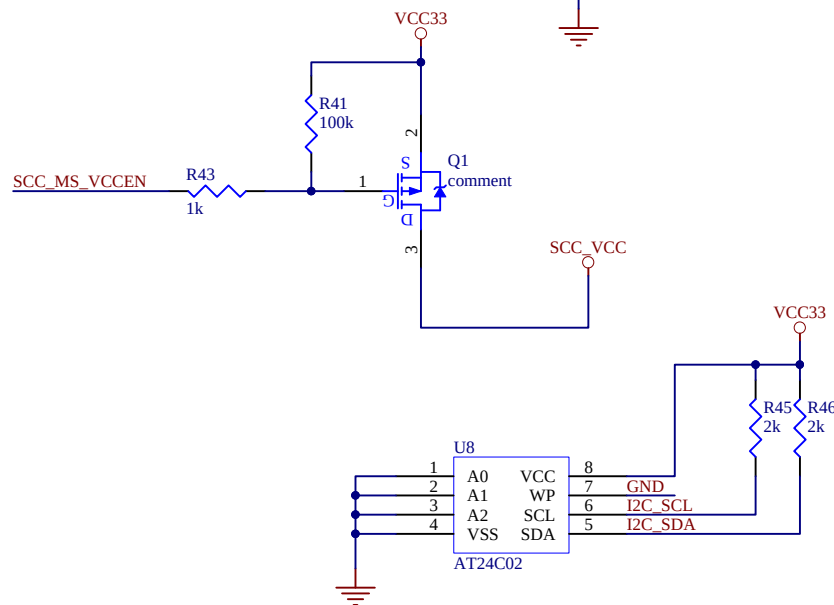
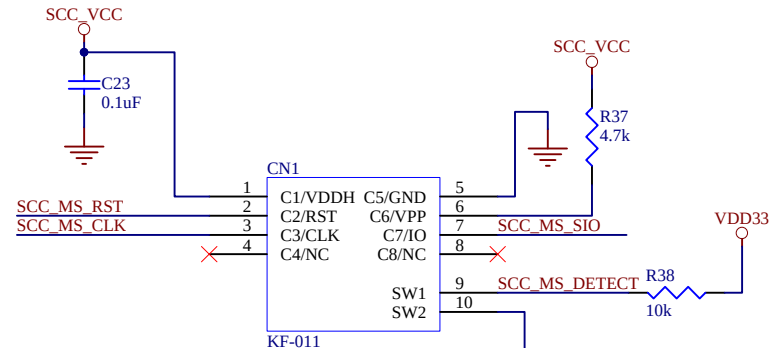
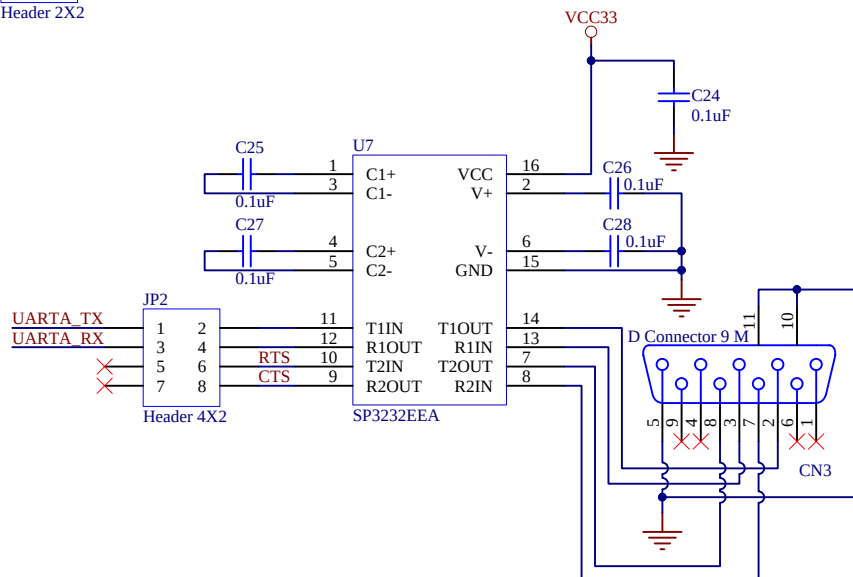
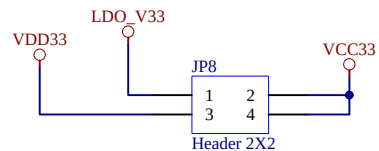
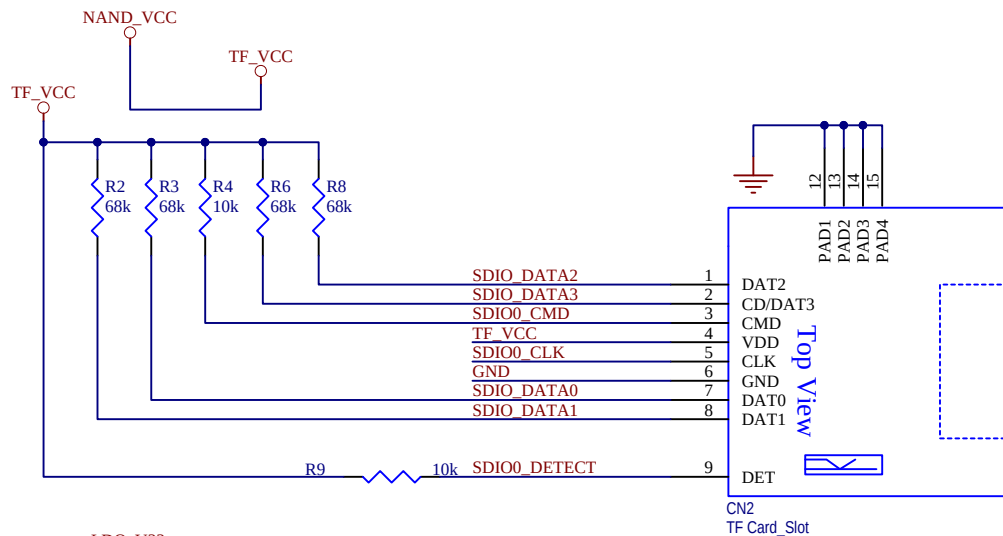
SLC NandFlash



Support NandFlash:

类型	厂家	型号	ECC bits
SLC	samsung	K9F1G08U0C/D	1bit/512
		K9F1G08U0A	1bit/512
		K9F2G08U0C	1bit/512
		K9F4G08U0D	1bit/512
	micron	MT29F1G08ABAEA	4bit/512
		MT29F2G08ABAEA	4bit/512
		MT29F4G08ABADA	4bit/512
		S34ML01G200	4bit/512
	spansion	S34ML02G100	1bit/512
		S34ML04G100	1bit/512
		S34ML02G200	4bit/512
		S34ML04G200	4bit/512
	toshiba	TC58NVG0S3HTA00	8bit/512
		TC58BVG0S3HTA00	8bit/512

 Shanghai AisinoChip Electronics Technology Co., Ltd.	Title: ACH512 V2
	Name: NandFlash
	Revision: 0.1
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 Shanghai AisinoChip Electronics Technology Co., Ltd.	Title: ACH512 V2
	Name: Peripheral
	Revision: 0.1
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3

4

A

B

C

D

			JP14				
GPIO45	NFM_CE1#	ADDR4	1	2	GPIO46	NFM_RB1#	ADDR5
GPIO43	UARTB_CTS	ADDR2	3	4	GPIO44	UARTB_RTS	ADDR3
GPIO37	SPIA_CS1	DATA8	5	6	GPIO36	I2C_SDA	SPIB_CS3#
GPIO39		DATA10	7	8	GPIO38	SPIB_CS1	DATA9
GPIO41	UARTB_TX	ADDR0	9	10	GPIO40		DATA11
GPIO14	SPIB_CS0#	CS0#	11	12	GPIO42	UARTB_RX	ADDR1
GPIO51	NFM_RB3#	DATA13	13	14	GPIO47		ADDR6
GPIO32	NFM_WE#	SDIO0_CMD	15	16			
GPIO34	NFM_WP#	SDIO0_WP#	17	18	✗ GPIO33	NFM_RE#	SDIO0_POWERen
GPIO28	NFM_CE0#	CS3#	19	20	GPIO35	I2C_SCL	SPIA_CS3#
GPIO26	SDIO_DATA6	NAND_DATA6	21	22	GPIO27	SDIO_DATA7	NAND_DATA7
GPIO24	SDIO_DATA4	NAND_DATA4	23	24	GPIO25	SDIO_DATA5	NAND_DATA5
GPIO22	SDIO_DATA2	NAND_DATA2	25	26	GPIO23	SDIO_DATA3	NAND_DATA3
GPIO20	SDIO_DATA0	NAND_DATA0	27	28	GPIO21	SDIO_DATA1	NAND_DATA1
GPIO18	SPIB_IO2	OE#	29	30	GPIO19	SPIB_IO3	WE#
GPIO16	SPIB_MOSI	ADDR14	31	32	GPIO17	SPIB_MISO	ADDR15
			Header 16X2				

Header 16X2

				JP13					
GPIO15	SPIB_CLK	ADDR13		1	2	✗	GPIO7	SPIA_IO3	ADDR19
GPIO6	SPIA_IO2	ADDR18		3	4		GPIO5	SPIA_MISO	
GPIO4	SPIA_MOSI			5	6	GPIO3	SPIA_CS0#	CS2#	
GPIO2	SPIA_CLK			7	8	GPIO58		ADDR12	
GPIO57		ADDR11		9	10	GPIO56		ADDR10	
GPIO31	NFM_ALE	SDIO0_CLK		11	12	GPIO30	NFM_CLE	SDIO0_DETECT	
GPIO29	NFM_RB0#			13	14	GPIO48	PWM2	ADDR7	REMAP
GPIO50	NFM_CE3#	RSTOUTN	DATA12	15	16	GPIO0	UARTA_RX		
GPIO1	UARTA_TX			17	18	GPIO52			DATA14
GPIO53		DATA15		19	20	GPIO54	NFM_CE2#	ADDR8	
GPIO55	NFM_RB2#	ADDR9		21	22				

Header 11X2

			JP12				
GPIO59	SCC_MS_VCCEN	PWM1	1	2	GPIO60	SCC_MS_DETECT	CS1#
GPIO61	SCC_MS_SIO		3	4	GPIO62	SCC_MS_CLK	SPIA_CS2#
GPIO63	SCC_MS_RST	SPIB_CS2#	5	6	GPIO13	ADDR16	BOOT
GPIO8	ADDR17		7	8	GPIO9	SWCLK	
GPIO10	ENB0#		9	10	GPIO11	SWDIO	
GPIO12	ENB1#		11	12	GPIO49	PWM0	CLKOUT
WAKE_UP			13	14			
			Header 7X2				

Header 7X2



 Shanghai AisinoChip Electronics Technology Co., Ltd.	Title: ACH512 V2
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